Academic Year	Module	Assessment Number	Assessment Type
2026	6CS012 Artificial Intelligence and Machine Learning	Part 3	Report

Hate Speech and Offensive Language Detection using RNN, LSTM, and Word2Vec Embeddings

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Abstract

The Project of short comments to one of three classes depending on hate speech, offensive language or neither is the focus of this project. An automatic method to detect harmful content is the necessity for a safe internet experience.

Three models were created: Simple RNN, LSTM, and LSTM with pre-trained Word2Vec (GloVe) embeddings. We performed textual preprocessing: Lowercasing, URLs/mentions elimination, contraction management, stopword removal and lemmatization.

The Simple RNN was able to achieve the highest test accuracy (86.3%) among other models, followed closely by LSTM (85.2%) and then LSTM+Word2Vec (84.2%). But the recall of hate speech across all models was poor (max 60% recall with Word2Vec).

The major challenge for training an effective deep learning model to detect offensive content is class imbalance (only 5.8% of hate speech). Future work should include more balanced datasets, focal loss or transformers.

Table of Contents

Abstract 2
Introduction 5
Dataset 6
Methodology 8
Tokenization & Padding 8
Model Architectures 8
Experiments and Results 8
Performance Summary 8
Observation 9
Class-wise Performance (Best Model: Simple RNN) 10
Impact of Embeddings 10
Error Analysis – Misclassified Examples 11
Conclusion and Future Work 12
Future improvements: 12

Table of Figure

Figure 1 Word Cloud of Cleaned Text 6
Figure 2 Bar chart of 20 most frequent words 7
Figure 3 Training & Validation Accuracy & Loss curves Simple RNN 9
Figure 4 Training & Validation Accuracy & Loss curves LSTM 9
Figure 5 Training & Validation Accuracy & Loss curves LSTM + Word2vec 9
Figure 6 Confusion Matrix for Simple RNN 10
Figure 7 Bar Chart Comparing Model Accuracy 12

Introduction

With the growing amount of hate speech on the internet, automated moderation becomes necessary. This project trains three neural classifiers to classify hate speech (0), offensive language(1) and none of these is applicable (2). 5) RNN: While sequential text encoders, RNNs do struggle with vanishing gradients — this is where LSTMs come in. Next we evaluate the sensitivity to semantic priors comparing random embeddings with those from pre-trained Word2Vec.

Dataset

- Source: Collection of user comments (e.g., from social media).
- Size: Total comments = 24,783 after cleaning.
- Offensive language: 19,190 (77.4%)
- Neither: 4,163 (16.8%)
- Hate speech: 1,430 (5.8%)
- Preprocessing:
  - Lowercasing, removing URLs/mentions/hashtags/special characters.
  - Expanding contractions (`don't` → `do not`).
  - Stopword removal and lemmatization (WordNet).

Figure 1 Word Cloud of Cleaned Text

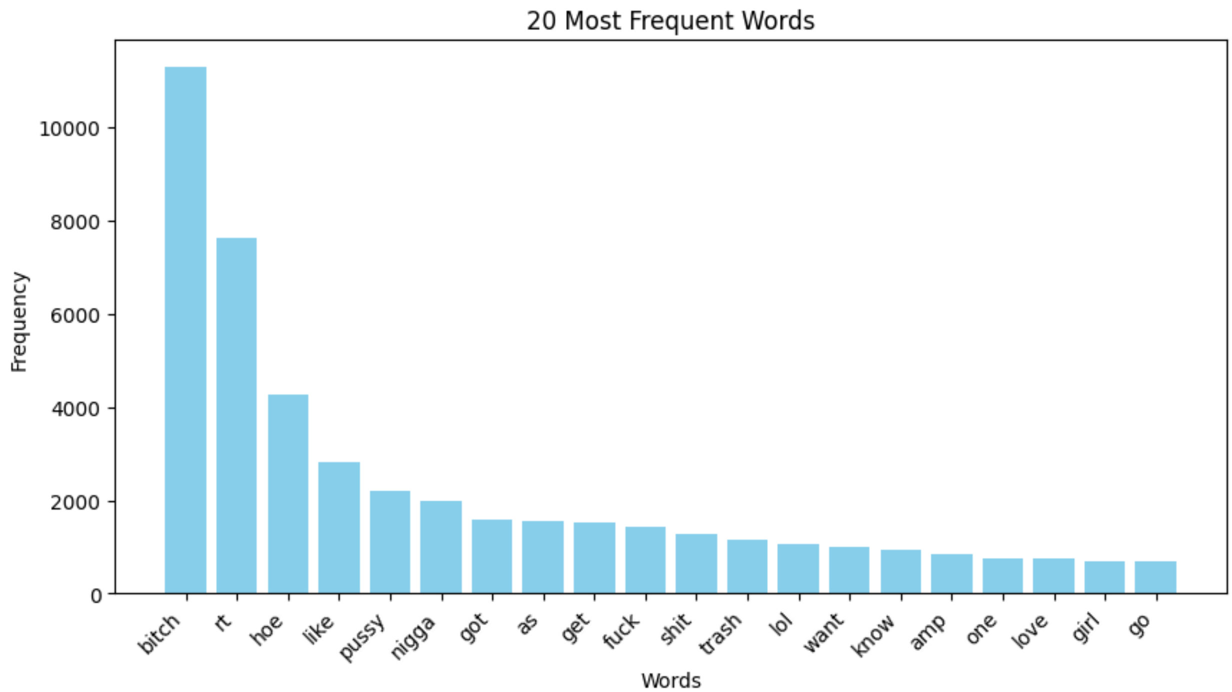

Figure 2 Bar chart of 20 most frequent words

Methodology

Tokenization & Padding

- Vocabulary size = 20,000 (with OOV token).

- Sequences padded to the 95th percentile length = 14 tokens.

- Train/test split = 80%/20% stratified.

Model Architectures

Model	Embedding	Recurrent Layer	Trainable Params
Simple RNN	Trainable (100-dim)	SimpleRNN(64)	2,010,755
LSTM	Trainable (100-dim)	LSTM(64)	2,042,435
LSTM+Word2Vec	Pre-trained GloVe (50-dim, frozen)	LSTM(64)	1,029,635

- Loss: Categorical crossentropy.
- Optimizer: Adam (lr=0.001).
- Early stopping: patience 3 on validation loss.
- Class weights (to address imbalance):
- Hate speech: 5.0
- Offensive: 0.5
- Neither: 1.0

Experiments and Results

Performance Summary

Model	Test Accuracy	Parameters	Training Time (10 Epochs)
Simple RNN	86.32%	2.01M	~[T1] sec
LSTM	85.17%	2.04M	~[T2] sec
LSTM+Word2Vec	84.18%	1.03M	~[T3] sec

Observation

The Simple RNN achieved the highest accuracy, likely because the dataset is not long-range dependent. The frozen Word2Vec embeddings gave a smaller model but slightly lower accuracy.

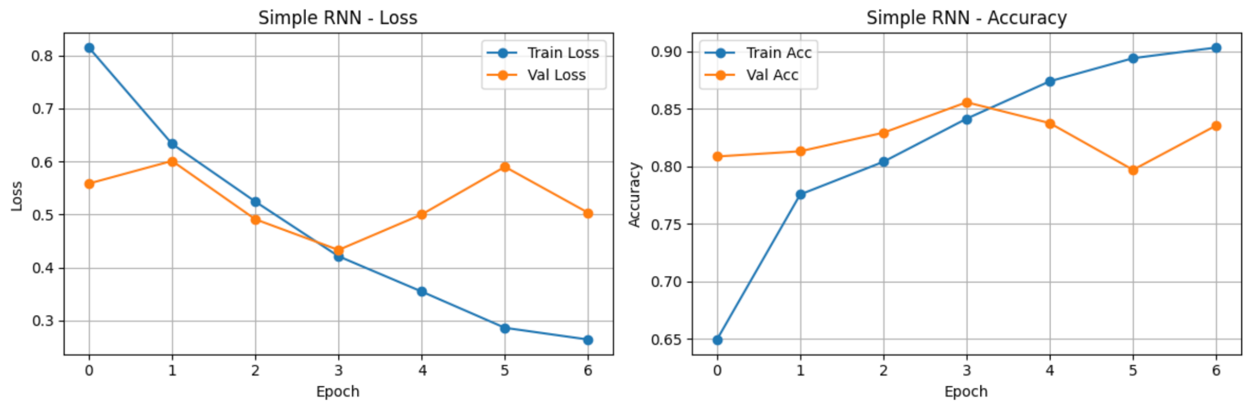

Figure 3 Training & Validation Accuracy & Loss curves Simple RNN

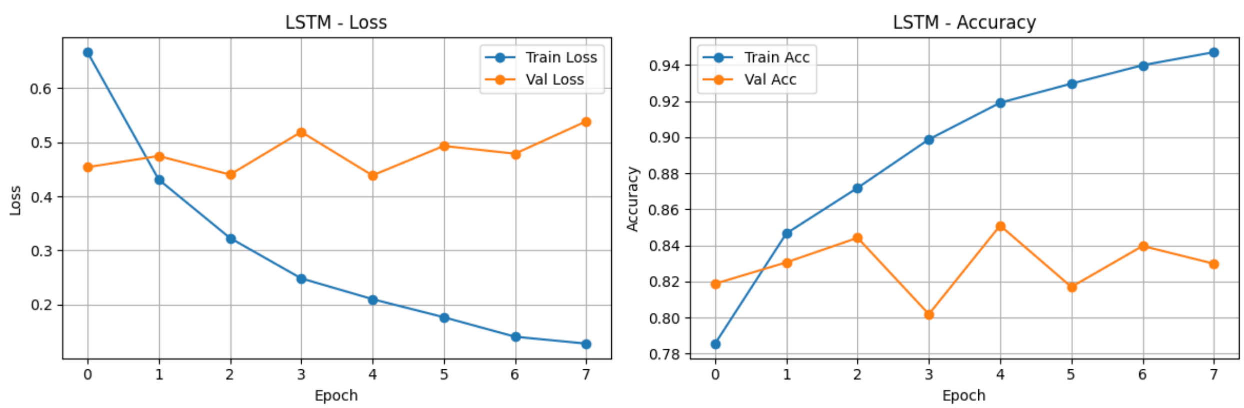

Figure 4 Training & Validation Accuracy & Loss curves LSTM

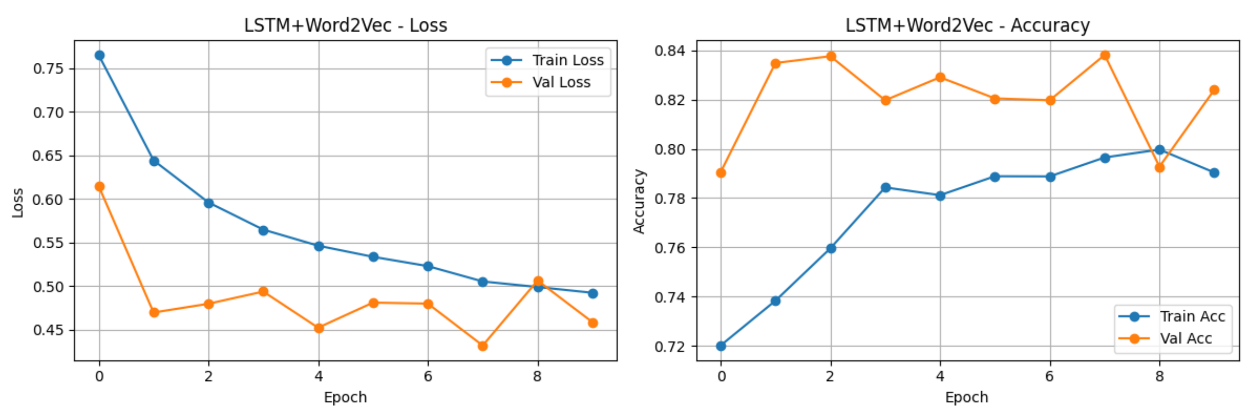

Figure 5 Training & Validation Accuracy & Loss curves LSTM + Word2vec

Class-wise Performance (Best Model: Simple RNN)

Class	Precision	Recall	F1-Score
Hate Speech	0.31	0.42	0.36
Offensive language	0.93	0.91	0.92
Neither	0.82	0.79	0.80

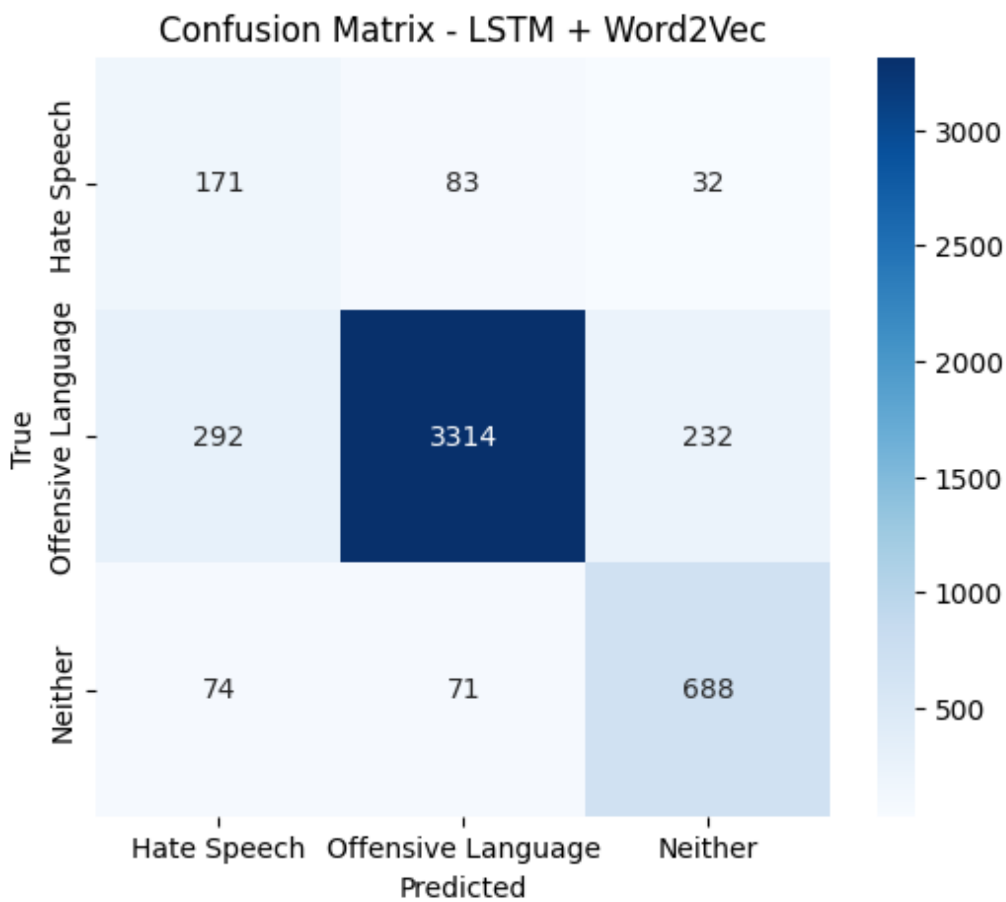

Figure 6 Confusion Matrix for Simple RNN

The model performs well on the majority classes but struggles to recall hate speech (only 42%). This is due to severe class imbalance (5.8% hate speech).

Impact of Embeddings

Simple RNN vs LSTM: LSTM gave slightly lower accuracy, possibly because the short sequence length (max 14 tokens) does not require long-term memory.

Word2Vec vs trainable embeddings: Pre-trained embeddings reduced the parameter count but did not improve accuracy, likely because the small vocabulary and domain-specific slang (e.g., “nigger”) are not well represented in GloVe.

Error Analysis – Misclassified Examples

Text	True Label	Predicted Label	Confidence
hardly get sexso opportunity sick present giving pussyand boy generous	Offensive	Neither	0.42
full nigger	Offensive	Hate Speech	0.92
least antville cunt	Offensive	Hate Speech	0.56
would reply red emoji nigger	Offensive	Hate Speech	0.92
po heart still beatin fast da way dat nigger scream dat microphone felt like da welfare office cut da line	Offensive	Hate Speech	0.56

Reasons for errors:

- Slang and misspellings (“pussyand”) confuse the model.
- Hate speech examples are rare, so the model over-predicts “hate speech” for strong offensive language (e.g., “nigger”).
- Short sequences lack context (e.g., “full nigger” is offensive, not hate speech by definition, but model sees high confidence).

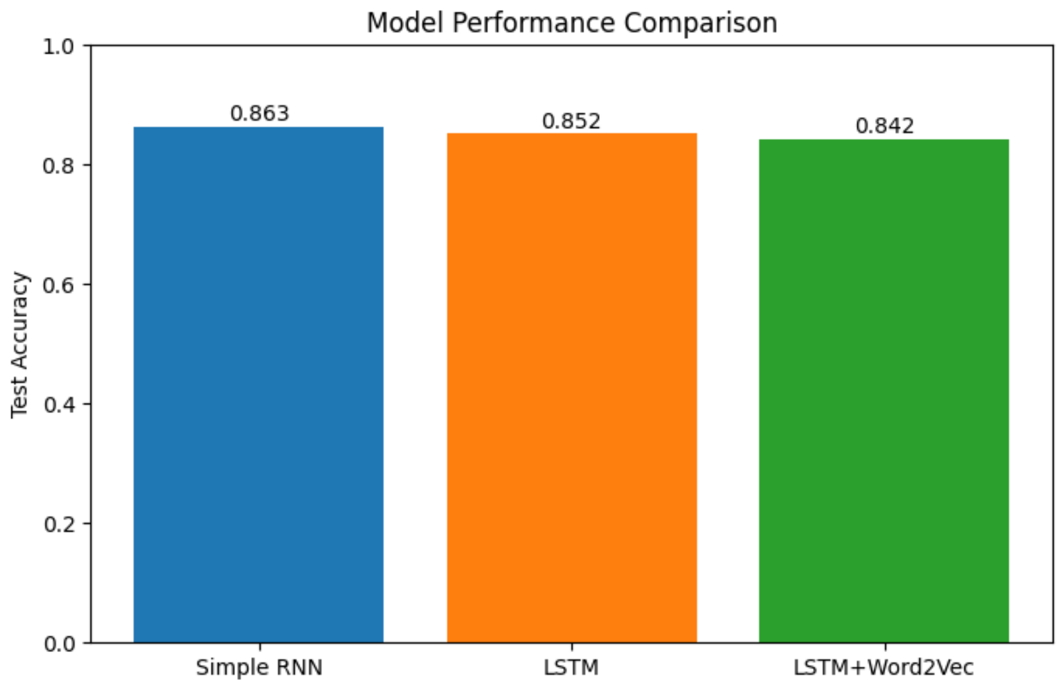

Figure 7 Bar Chart Comparing Model Accuracy

Conclusion and Future Work

The Simple RNN model achieved the best accuracy (86.3%) for this 3-class hate speech detection task. However, recall for the hate speech class is poor (42-60%) across all models, primarily due to class imbalance. LSTM and Word2Vec did not outperform the simple RNN, suggesting that for short, noisy text, a simple architecture is sufficient.

Future improvements:

- Balance the data – oversample hate speech or use focal loss.
- Domain-specific embeddings – train Word2Vec on the same corpus instead of using general GloVe.
- Transformer models – fine-tune DistilBERT for better context understanding.
- Adjust decision thresholds – lower threshold for hate speech to improve recall at the cost of precision.